

An Extra Residual Spectral Point on the Barlow–Perkins Sierpinski Lattice

Candidate counterexample for arXiv:1910.01771

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Status: candidate counterexample, likely valid. The Barlow–Perkins question is answered negatively, subject to human review.

1 Source question

Shiping Cao, Yiqi Huang, Hua Qiu, Robert S. Strichartz, and Xiaohan Zhu study the graph Laplacian on infinite Sierpinski lattices in [1]. In Section 3.3 (arXiv:1910.01771, PDF page 14), they write the Barlow–Perkins lattice $\widetilde{SG}_{BP}$ as the union of two boundary-point Sierpinski half-lattices and observe that reflection splits its Laplacian into a Neumann and a Dirichlet half. After proving the $1 < p < \infty$ spectral classification, they ask:

On the other hand, the argument in Proposition 3.4, showing that the eigenfunctions are not bounded, relies heavily on (3.2). It is not clear how to overcome this, and we do not know whether the same result of Theorem 3.3 (b) holds on $\widetilde{SG}_{BP}$.

boundary, as the approach for Proposition 3.4 is no longer available in this situation. It seems possible that the eigenvalues for $\Delta : \ell^\infty(\widetilde{\mathcal{SG}}) \rightarrow \ell^\infty(\widetilde{\mathcal{SG}})$ depend on the generating sequence ω of the lattice. We leave the problem for further study.

Problem. Describe all the eigenvalues of $\Delta : \ell^\infty(\widetilde{\mathcal{SG}}) \rightarrow \ell^\infty(\widetilde{\mathcal{SG}})$. In particular, whether the set of ℓ^∞ -eigenvalues depends on the generating sequence $\omega \in \{0, 1, 2\}^\infty$?

3.3. Infinite Sierpinski lattice of Barlow and Perkins. At the end of this section, we point out that there is another important infinite Sierpinski lattice in history, introduced by M.T. Barlow and E.A. Perkins [4]. We write $\widetilde{\mathcal{SG}}_{BP}$ for this lattice, see Figure 7.

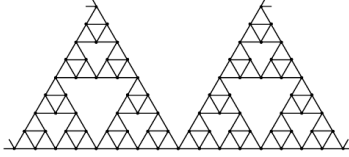


FIGURE 7. The infinite Sierpinski lattice $\widetilde{\mathcal{SG}}_{BP}$ of Barlow and Perkins.

This example is not an infinite Sierpinski lattice considered above, it is the union of two copies of $\widetilde{\mathcal{SG}}$ which are joined at the boundary. By symmetry, one can easily see that the Laplacian on this lattice is equivalent to a direct sum of Δ and Δ_D , where Δ is the Neumann Laplacian we consider above, and Δ_D is the Dirichlet Laplacian on $\widetilde{\mathcal{SG}}$ with a boundary point.

The argument in Section 2 applies here without difficulty. In addition, one can modify the argument of Proposition 3.4 to show that the ℓ^p -eigenvalues of Δ on $\widetilde{\mathcal{SG}}_{BP}$ is still $\Sigma_5 \cup \Sigma_6$. In fact, we have to drop (3.2) in the argument since we no longer have the Neumann boundary condition on a half of $\widetilde{\mathcal{SG}}_{BP}$, but still we can see if $\lambda \in \mathcal{J} \setminus (\{0\} \cup \Sigma_4 \cup \Sigma_5)$ and f is a non-zero λ -eigenfunction, $|f|$ is bounded below on infinitely many points of the form $F_0^{-m} F_i^m q_j$, with $i \in \{1, 2\}$ and $j \neq i$. We leave the detail to readers.

Theorem 3.10. For $1 < p < \infty$, $\Delta : \ell^p(\widetilde{\mathcal{SG}}_{BP}) \rightarrow \ell^p(\widetilde{\mathcal{SG}}_{BP})$ has point spectrum $\sigma_p(\Delta) = \Sigma_5 \cup \Sigma_6$, and continuous spectrum $\sigma_c(\Delta) = \mathcal{J} \setminus \Sigma_5$. There is no residue spectrum.

On the other hand, the argument in Proposition 3.4 showing that the eigenfunctions are not bounded, relies heavily on (3.2). It is not clear how to overcome this, and we do not know whether the same result of Theorem 3.3 (b) holds on $\widetilde{\mathcal{SG}}_{BP}$.

Figure 1: The Barlow–Perkins discussion and question in arXiv:1910.01771, PDF page 14.

Theorem 3.3(b) says, for a one-sided lattice with a boundary point, that the ℓ^1 residual spectrum is $\{0\} \cup \Sigma_4$ and that

$$\mathcal{J} \setminus (\Sigma_4 \cup \Sigma_5 \cup \{0\})$$

is continuous spectrum. Here

$$R(t) = t(5 - t), \quad \Sigma_4 = \bigcup_{m \geq 0} R^{-m}\{4\}, \quad \Sigma_5 = \bigcup_{m \geq 0} R^{-m}\{5\},$$

and

$$\Sigma_6 = \{6\} \cup \bigcup_{m \geq 0} R^{-m}\{3\}.$$

As in the source, a “ λ -eigenfunction” satisfies $-\Delta f = \lambda f$.

2 Proof intuition

The missing odd-reflection half is a Dirichlet problem. We choose the period-two point

$$a = 3 - \sqrt{3}, \quad R(a) = 3 + \sqrt{3} =: b, \quad R(b) = a.$$

Two steps of spectral decimation therefore repeat exactly. Symmetric boundary data on the Dirichlet half are multiplied by -3 when transported two levels inward, so prescribing outer data $(-1/3)^n(0, 1, 1)$ on a level- $2n$ gasket makes all finite approximants consistent.

Consistency alone does not prove boundedness: individual extension coefficients are larger than one. The key certificate is a positive quadratic form P for which every one of the nine possible two-level child-extension matrices has norm at most 3. The prefactor 3^{-n} cancels this worst-case growth exactly. The consistent finite solutions therefore converge to a nonzero bounded Dirichlet eigenfunction. Odd reflection across the gluing vertex gives a bounded a -eigenfunction on $\widetilde{SG}_{BP}$. Duality then puts a in the ℓ^1 residual spectrum, contrary to the proposed extension of Theorem 3.3(b).

3 The two-level extension certificate

For a finite gasket cell, order the three boundary values as $v = (v_0, v_1, v_2)^T$. Proposition 2.3 of [1] gives the usual spectral decimation extension rule. If the fine eigenvalue is t , the value at the midpoint of the edge joining vertices i, j is

$$\frac{(4 - t)(v_i + v_j) + 2v_k}{(2 - t)(5 - t)}, \tag{1}$$

where $\{i, j, k\} = \{0, 1, 2\}$.

For $t = a$ and $t = b$, the coefficient of $v_i + v_j$ in (1) is one, while the coefficient of v_k is respectively

$$c_a = \sqrt{3} - 1, \quad c_b = -(1 + \sqrt{3}). \tag{2}$$

For a scalar c , define the three child-boundary matrices

$$L_0(c) = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & c \\ 1 & c & 1 \end{pmatrix}, \quad L_1(c) = \begin{pmatrix} 1 & 1 & c \\ 0 & 1 & 0 \\ c & 1 & 1 \end{pmatrix}, \quad L_2(c) = \begin{pmatrix} 1 & c & 1 \\ c & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3}$$

Thus a two-level child has boundary vector

$$M_{ij}v, \quad M_{ij} = L_j(c_a)L_i(c_b), \quad 0 \leq i, j \leq 2. \tag{4}$$

Lemma 1 (Common quadratic bound). *Let J denote the 3×3 all-ones matrix and put*

$$P = I - \frac{\sqrt{3}}{6}J.$$

Then P is positive definite and, for all $0 \leq i, j \leq 2$,

$$M_{ij}^T P M_{ij} \leq 9P. \quad (5)$$

Moreover, for $v_ = (0, 1, 1)^T$,*

$$M_{00}v_* = -3v_*. \quad (6)$$

Proof. The eigenvalues of P are $1, 1, 1 - \sqrt{3}/2$, so $P > 0$. The matrices in (4) are equivariant under simultaneous permutations of the three coordinates, and P is permutation invariant. Hence there are only two cases: $i = j$ and $i \neq j$.

For representatives M_{00} and M_{01} , direct multiplication gives

$$9P - M_{00}^T P M_{00} = \begin{pmatrix} 6 - 2\sqrt{3} & 0 & 0 \\ 0 & 4 & -4 \\ 0 & -4 & 4 \end{pmatrix} \geq 0, \quad (7)$$

and

$$9P - M_{01}^T P M_{01} = \begin{pmatrix} 6 - 2\sqrt{3} & (3 - \sqrt{3})/2 & -\sqrt{3} \\ (3 - \sqrt{3})/2 & 4 - \sqrt{3} & (-5 + \sqrt{3})/2 \\ -\sqrt{3} & (-5 + \sqrt{3})/2 & 4 \end{pmatrix}. \quad (8)$$

The leading principal minors of (8) are

$$6 - 2\sqrt{3}, \quad \frac{54 - 25\sqrt{3}}{2}, \quad 9(3 - \sqrt{3}),$$

all strictly positive. Thus (8) is positive definite and (5) follows. Finally, (3)–(4) give (6) directly. $\square$

Write $\|v\|_P = (v^T P v)^{1/2}$. Lemma 1 says

$$\|M_{ij}v\|_P \leq 3\|v\|_P. \quad (9)$$

4 Construction of the bounded eigenfunction

Theorem 2. *Let $a = 3 - \sqrt{3}$. Then a is an ℓ^∞ eigenvalue of the Barlow–Perkins Sierpinski lattice Laplacian. Consequently, a belongs to the residual spectrum of the Laplacian on $\ell^1(\widetilde{SG}_{BP})$. In particular, the analogue of Theorem 3.3(b) in [1] is false on $\widetilde{SG}_{BP}$.*

Proof. We first work on one boundary-point half with Dirichlet value zero at its boundary vertex. Let G_{2n} be the nested level- $2n$ finite gasket in that half. At its three outer corners prescribe

$$d_n v_*, \quad d_n = \left(-\frac{1}{3}\right)^n, \quad v_* = (0, 1, 1)^T. \quad (10)$$

Starting with this outer boundary vector, extend through n pairs of levels, first with the b rule and then with the a rule in each pair. Formula (1) shows that this produces a function f_n satisfying $-\Delta f_n = a f_n$ at every vertex of G_{2n} except its three outer corners.

The distinguished child obtained by choosing the 0-cell twice has boundary vector

$$d_n M_{00} v_* = -3d_n v_* = d_{n-1} v_*$$

by (6). Repeating inside that child shows that f_n restricts to f_{n-1} on the naturally embedded G_{2n-2} . The finite functions are therefore consistent.

Every vertex value of f_n is a coordinate of

$$d_n M_{i_n j_n} \cdots M_{i_1 j_1} v_* \tag{11}$$

for some sequence of n two-level children. By (9), the P -norm of (11) is at most

$$|d_n| 3^n \|v_*\|_P = \|v_*\|_P. \tag{12}$$

Since $P > 0$, Euclidean coordinate norm is bounded by a fixed multiple of $\|\cdot\|_P$. Thus all f_n are bounded by one constant independent of n .

The consistent union $f_D = \bigcup_n f_n$ is consequently a bounded function on the infinite Dirichlet half. Every nonboundary vertex is an interior vertex of some G_{2n} , so $-\Delta_D f_D = a f_D$ everywhere. It is nonzero: on the two non-Dirichlet vertices of the initial cell its values are both one.

Now use the reflection of $\widehat{SG}_{BP}$ exchanging its two halves. Put f_D on one half, $-f_D$ on the other, and zero at the common vertex. The eigenvalue equation holds away from the common vertex by construction. At the common vertex, the four neighboring values cancel in reflected pairs, so the equation holds there as well. This gives a nonzero bounded a -eigenfunction on $\widehat{SG}_{BP}$.

The orbit $a \mapsto b \mapsto a$ is bounded, hence $a \in \mathcal{J}$. It never meets 0, 3, 4, 5, or 6, so

$$a \notin \{0\} \cup \Sigma_4 \cup \Sigma_5 \cup \Sigma_6. \tag{13}$$

In particular, Theorem 3.10 of [1] at $p = 2$ shows that a is not an ℓ^2 eigenvalue; since $\ell^1 \subset \ell^2$, it is not an ℓ^1 eigenvalue either.

On the other hand, the bounded eigenfunction just constructed belongs to the kernel of the adjoint of $a + \Delta$ acting on ℓ^1 . Hence the range of $a + \Delta$ on ℓ^1 is not dense. Since $a + \Delta$ is injective there, a is a residual spectral point. Theorem 3.3(b) would instead place (13) in the continuous spectrum, proving the claimed negative answer. $\square$

5 Verification, limitations, and novelty check

The exact matrix certificate is independently reproduced by `code/verify_matrix_certificate.py`. It checks all nine inequalities in (5), positivity of P , and (6) symbolically over $\mathbb{Q}(\sqrt{3})$. The auxiliary `code/finite_level_probe.py` builds finite gasket graphs and solves the boundary problems numerically. Levels 1 through 8 (up to 9843 vertices) retain the inner values (1, 1) and have observed supremum $2/\sqrt{3}$. These computations are checks, not ingredients of the proof.

This counterexample fully answers the specific yes/no Barlow–Perkins question negatively. It does not classify the entire ℓ^1 spectrum of $\widehat{SG}_{BP}$, nor does it answer the broader preceding problem about all ℓ^∞ eigenvalues of arbitrary boundaryless Sierpinski lattices or their dependence on the generating word.

A bounded literature search checked arXiv:1910.01771, the exact sentence “we do not know whether the same result”, and combinations of “Barlow–Perkins”, “Sierpinski lattice”, “ ℓ^1 spectrum”, “ ℓ^∞ eigenvalue”, “generating sequence”, and “ $3 - \sqrt{3}$ ”. Searches also checked later arXiv results on Sierpinski lattice spectra and spectral dimension. No later source answering this question or giving the eigenvalue above was located. This supports, but does not certify, novelty.

Human-review recommendation. Review as a likely valid full negative answer to the Barlow–Perkins question. The highest-value checks are the orientation/order of the two spectral decimation extensions in (4), consistency of the nested finite gaskets, and the final ℓ^∞ – ℓ^1 adjoint argument.

References

- [1] S. Cao, Y. Huang, H. Qiu, R. S. Strichartz, and X. Zhu, *Spectral analysis beyond ℓ^2 on Sierpinski lattices*, arXiv:1910.01771, version 2 (2021), <https://arxiv.org/abs/1910.01771>.